

Erweiterte_Ergebnisse_BA

Setup & Daten laden

```
options(repos = c(CRAN = "https://cran.rstudio.com/"))  
if (!require("irr")) install.packages("irr")
```

Lade nötiges Paket: irr

Warning: Paket 'irr' wurde unter R Version 4.4.2 erstellt

Lade nötiges Paket: lpSolve

Warning: Paket 'lpSolve' wurde unter R Version 4.4.2 erstellt

```
if (!require("psych")) install.packages("psych")
```

Lade nötiges Paket: psych

Warning: Paket 'psych' wurde unter R Version 4.4.3 erstellt

```
library(irr)  
library(psych)
```

1. Daten laden

```
all_files <- list.files(".", pattern="*.csv", full.names=TRUE)
person_files <- all_files[!grepl("results_run", all_files)]
person_files <- person_files[!grepl("Testset", person_files)]

all_data <- do.call(rbind, lapply(person_files, function(f){
  read.csv(f, stringsAsFactors=FALSE)
}))
```

1.2 KI-Modell-Ergebnisse Zero-Shot & Lora laden

```
zs <- read.csv("results_run_zs_1.csv", stringsAsFactors=FALSE)
lora <- read.csv("results_run_lora_1.csv", stringsAsFactors=FALSE)
```

2. Datenaufbereitung

Prioritäten numerisch codieren

Blocker=5, Critical=4, Major=3, Minor=2, Trivial=1 (höherer Wert = höhere Priorität)

```
prio_map <- c(Blocker=5, Critical=4, Major=3, Minor=2, Trivial=1)

all_data$priority_num <- prio_map[all_data$priority]
zs$pred_num <- prio_map[zs$predicted_priority]
lora$pred_num <- prio_map[lora$predicted_priority]
zs$jira_num <- prio_map[zs$original_priority]
```

2.2 Personen-IDs zuweisen mit Gruppendesign: Gruppe A = Kontrollgruppe (keine KI); Gruppe B = Human-in-the-Loop mit Zero-Shot-Modell; Gruppe C = Human-in-the-Loop mit LoRA-Modell

```
file_to_person <- c(
  "christian_zotter_developer_a_d8ace3eb48824b24ba40fcce652985e8"= "A1",
  "ferdi_sulejman_projectmanager_a_ccb85493a1604cb2b7f90d5ff0e8a7f8"= "A2",
  "stefan_stojadinovic_other_a_17d382d2057b44c192952961f56c9b6f"= "A3",
  "aida_omic_developer_b_cfc39c24bd96401f9926dfab1d583d30"= "B1",
  "claudia_schrammel_projectmanager_b_99e0f42f9b99455199ae33ae83f28d61"= "B2",
  "kristian_karanitsch_other_b_91d8d372f4f54e83865b7cd94f19dc4d"= "B3",
  "ermin_hasic_projectmanager_c_ad3fe0a794944357b46aa3a5346a18b5"= "C1",
  "robert_hinternberger_developer_c_57c441ce044d4ceb9ce0792a860cb177"= "C2",
  "lukas_schmidberger_other_c_19d39e942dca4f568df0bffe4f8c0b"= "C3"
)
```

```
all_data <- do.call(rbind, lapply(person_files, function(f) {
  df <- read.csv(f, stringsAsFactors = FALSE)
  basename_f <- tools::file_path_sans_ext(basename(f))
  df$person_id <- file_to_person[basename_f]
  df
}))

prio_map <- c(Blocker=5, Critical=4, Major=3, Minor=2, Trivial=1)
all_data$priority_num <- prio_map[all_data$priority]
```

2.3 Issues sortieren und Pivot-Tabelle aufbauen Zeilen = Issues (n=20), Spalten = Personen + KI-Modelle

```
issues <- sort(unique(all_data$story_id))
alle_personen <- c("A1","A2","A3","B1","B2","B3","C1","C2","C3")
pivot <- data.frame(story_id = issues)

for(p in alle_personen){
  sub <- all_data[all_data$person_id == p, c("story_id","priority_num")]
  sub <- sub[match(issues, sub$story_id), ]
  pivot[[p]] <- sub$priority_num
}
```

2.4 Jira-Originalprioritäten, Zero-Shot und LoRA anhängen

```
zs_match <- zs[match(issues, zs$key), ]
lora_match <- lora[match(issues, lora$key), ]

pivot$Jira <- zs_match$jira_num
pivot$ZeroShot <- zs_match$pred_num
pivot$LoRA <- lora_match$pred_num
```

2.5 Gruppenmittelwert Gruppe A berechnen

```
pivot$A_mean <- rowMeans(pivot[, c("A1","A2","A3")], na.rm=TRUE)

spearman_ci <- function(x, y, label_x, label_y, n_boot = 2000) {
  res <- cor.test(x, y, method = "spearman", exact = FALSE)

  set.seed(42)
```

```

boot_rhos <- replicate(n_boot, {
  idx <- sample(length(x), replace = TRUE)
  cor(x[idx], y[idx], method = "spearman")
})
ci_lo <- quantile(boot_rhos, 0.025, na.rm = TRUE)
ci_hi <- quantile(boot_rhos, 0.975, na.rm = TRUE)

cat(sprintf(
  "%s vs %s : rho= %.3f  p= %.4f  95%-KI [%.3f, %.3f]\n",
  label_x, label_y,
  round(res$estimate, 3),
  round(res$p.value, 4),
  round(ci_lo, 3),
  round(ci_hi, 3)
))}

```

K4.1: RQ1: Übereinstimmung KI-Modelle mit Jira-Prioritäten; Methode: Spearman-Rangkorrelation

```
spearman_ci(pivot$ZeroShot, pivot$Jira, "ZeroShot", "Jira")
```

ZeroShot vs Jira : rho= 0.629 p= 0.0030 95%-KI [0.343, 0.806]

```
spearman_ci(pivot$LoRA, pivot$Jira, "LoRA", "Jira")
```

LoRA vs Jira : rho= 0.711 p= 0.0004 95%-KI [0.400, 0.889]

```
spearman_ci(pivot$ZeroShot, pivot$LoRA, "ZeroShot", "LoRA")
```

ZeroShot vs LoRA : rho= 0.860 p= 0.0000 95%-KI [0.676, 0.986]

K4.2.1: RQ2: Vergleich Zero-Shot vs. LoRA Methode: Deskriptive Kriterienbewertungsanalyse

```
cat("ZERO-SHOT KRITERIEN\n")
```

ZERO-SHOT KRITERIEN

```

for (crit in c("business value", "urgency", "impact",
              "risk", "estimated effort")) {
  cat("\n", crit, "\n")
  print(table(zs[[crit]]))
  cat("Prozent:\n")
  print(round(prop.table(table(zs[[crit]])) * 100, 1))
}

```

```

business value
< table of extent 0 >
Prozent:
numeric(0)

```

```

urgency

  High    Low Medium
    8     11      1
Prozent:

```

```

  High    Low Medium
   40     55      5

```

```

impact

  High    Low Medium
    6     11      3
Prozent:

```

```

  High    Low Medium
   30     55     15

```

```

risk

High Low
  7   13
Prozent:

```

```

High Low
  35   65

```

```

estimated effort

```

```
< table of extent 0 >
Prozent:
numeric(0)
```

```
cat("\nLORA KRITERIEN\n")
```

LORA KRITERIEN

```
for (crit in c("business value", "urgency", "impact",
               "risk", "estimated effort")) {
  cat("\n", crit, "\n")
  print(table(lora[[crit]]))
  cat("Prozent:\n")
  print(round(prop.table(table(lora[[crit]])) * 100, 1))
}
```

```
business value
< table of extent 0 >
Prozent:
numeric(0)
```

```
urgency
```

```
High Low
  11   9
```

```
Prozent:
```

```
High Low
  55  45
```

```
impact
```

```
High Low Medium
   8   9     3
```

```
Prozent:
```

```
High Low Medium
  40  45    15
```

risk

High	Low	Medium
10	9	1

Prozent:

High	Low	Medium
50	45	5

estimated effort

< table of extent 0 >

Prozent:

numeric(0)

K4.3: RQ3: KI-Modelle vs. Menschliche Expert*innen

Spearman: Gruppe A vs. Jira / Zero-Shot / LoRA Methode: Spearman-Rangkorrelation

```
cat("\nSPEARMAN: Gruppe A vs Jira / ZeroShot / LoRA:\n")
```

SPEARMAN: Gruppe A vs Jira / ZeroShot / LoRA:

```
for(p in c("A1","A2","A3","A_mean")){
  for(comp in c("Jira","ZeroShot","LoRA")){
    res <- cor.test(pivot[[p]], pivot[[comp]], method="spearman", exact=FALSE)
    cat(p, "vs", comp, ": rho=", round(res$estimate,3),
        " p=", round(res$p.value,4), "\n")
  }
}
```

```
A1 vs Jira : rho= 0.453 p= 0.0448
A1 vs ZeroShot : rho= 0.221 p= 0.3497
A1 vs LoRA : rho= 0.45 p= 0.0464
A2 vs Jira : rho= 0.465 p= 0.0388
A2 vs ZeroShot : rho= 0.438 p= 0.0534
A2 vs LoRA : rho= 0.617 p= 0.0038
A3 vs Jira : rho= 0.355 p= 0.125
A3 vs ZeroShot : rho= 0.126 p= 0.597
A3 vs LoRA : rho= 0.365 p= 0.1138
A_mean vs Jira : rho= 0.488 p= 0.0291
A_mean vs ZeroShot : rho= 0.337 p= 0.1459
A_mean vs LoRA : rho= 0.56 p= 0.0102
```

K4.3.1: Innerhalb Gruppe A - Paarweise Korrelation

```
cat("\nPaarweise Korrelationen innerhalb Gruppe A: \n")
```

Paarweise Korrelationen innerhalb Gruppe A:

```
for(pair in list(c("A1","A2"), c("A1","A3"), c("A2","A3"))){  
  res <- cor.test(pivot[[pair[1]]], pivot[[pair[2]]],  
                 method="spearman", exact=FALSE)  
  cat(pair[1], "vs", pair[2], ": rho=", round(res$estimate,3),  
      " p=", round(res$p.value,4), "\n")  
}
```

```
A1 vs A2 : rho= 0.54  p= 0.0139  
A1 vs A3 : rho= 0.335 p= 0.1486  
A2 vs A3 : rho= 0.704 p= 5e-04
```

K4.3.1: Richtungsanalyse: Gruppe A (Mittelwert) vs. KI zeigt ob Mensch systematisch höher bzw. niedriger als KI bewertet

```
for (ki in c("ZeroShot", "LoRA")) {  
  diff <- pivot$A_mean - pivot[[ki]]  
  cat(sprintf("A_mean minus %s:\n", ki))  
  cat("Positiv (Mensch höher): ", sum(diff > 0, na.rm = TRUE), "\n")  
  cat("Negativ (Mensch niedriger):", sum(diff < 0, na.rm = TRUE), "\n")  
  cat("Gleich: ", sum(diff == 0, na.rm = TRUE), "\n\n")  
}
```

```
A_mean minus ZeroShot:  
Positiv (Mensch höher): 4  
Negativ (Mensch niedriger): 9  
Gleich: 7
```

```
A_mean minus LoRA:  
Positiv (Mensch höher): 2  
Negativ (Mensch niedriger): 11  
Gleich: 7
```

K4.1: Abweichungsanalyse von Jira (± 0 , ± 1 , ± 2 +))

```

for (p in c("A1", "A2", "A3", "ZeroShot", "LoRA")) {
  abw <- abs(pivot[[p]] - pivot$Jira)
  kat <- ifelse(abw == 0, "exakt",
               ifelse(abw == 1, "±1", "±2+"))
  cat(p, ":\n")
  print(table(kat))
  cat("\n")
}

```

```

A1 :
kat
  ±1  ±2+ exakt
  12   4     4

```

```

A2 :
kat
  ±1  ±2+ exakt
   8   6     6

```

```

A3 :
kat
  ±1  ±2+ exakt
  11   7     2

```

```

ZeroShot :
kat
  ±1  ±2+ exakt
   9   4     7

```

```

LoRA :
kat
  ±1  ±2+ exakt
  11   1     8

```

K4.3.2: Kriterienbewertungen: Gruppe A vs. Zero-Shot vs. LoRA Methode: Deskriptive Kriterienbewertungsanalyse

```

group_a <- all_data[all_data$group == "A", ]

for (crit in c("business value", "urgency", "impact",
               "risk", "estimated effort")) {

```

```

cat("", crit, "\n")
print(table(group_a[[crit]]))
cat("Prozent:\n")
print(round(prop.table(table(group_a[[crit]])) * 100, 1))
cat("\n")
}

```

```

business value
< table of extent 0 >
Prozent:
numeric(0)

```

```

urgency

  High    Low Medium
    12     28     20
Prozent:

```

```

  High    Low Medium
 20.0   46.7   33.3

```

```

impact

  High    Low Medium
    13     27     20
Prozent:

```

```

  High    Low Medium
 21.7   45.0   33.3

```

```

risk

  High    Low Medium
     6    41     13
Prozent:

```

```

  High    Low Medium
 10.0   68.3   21.7

```

```

estimated effort
< table of extent 0 >
Prozent:

```

```
numeric(0)
```

K4.4: RQ4: Auswirkungen der KI-Unterstützung auf den Prozess

K4.4.1: Zeitaufwandsmessung Methode: Zeitaufwandsmessung; decision_time_s = Zeit bis zur Prioritätsentscheidung total_time_s = Gesamtzeit inkl. Begründung justif_time_s = Differenz = Kriterien + Begründungszeit

```
cat("4.4.1 ZEITAUFWAND PRO GRUPPE\n\n")
```

4.4.1 ZEITAUFWAND PRO GRUPPE

```
for (g in c("A", "B", "C")) {  
  sub <- all_data[all_data$group == g, ]  
  cat(sprintf("Gruppe %s:\n", g))  
  cat("  Ø Entscheidungszeit (s):",  
      round(mean(sub$decision_time_s, na.rm = TRUE), 1), "\n")  
  cat("  Ø Gesamtzeit (s):      ",  
      round(mean(sub$total_time_s, na.rm = TRUE), 1), "\n")  
  cat("  Ø Begründungszeit (s):  ",  
      round(mean(sub$total_time_s - sub$decision_time_s,  
                  na.rm = TRUE), 1), "\n")  
  cat("  SD Entscheidungszeit:   ",  
      round(sd(sub$decision_time_s, na.rm = TRUE), 1), "\n")  
  cat("  SD Gesamtzeit:          ",  
      round(sd(sub$total_time_s, na.rm = TRUE), 1), "\n\n")  
}
```

Gruppe A:

```
Ø Entscheidungszeit (s): 38.8  
Ø Gesamtzeit (s):      117.5  
Ø Begründungszeit (s):  78.8  
SD Entscheidungszeit:   21.3  
SD Gesamtzeit:          56.2
```

Gruppe B:

```
Ø Entscheidungszeit (s): 48.7  
Ø Gesamtzeit (s):      101.6  
Ø Begründungszeit (s):  52.9  
SD Entscheidungszeit:   27.4  
SD Gesamtzeit:          67.2
```

Gruppe C:

```
Ø Entscheidungszeit (s): 58
Ø Gesamtzeit (s):      134.6
Ø Begründungszeit (s): 76.6
SD Entscheidungszeit:   33.9
SD Gesamtzeit:          76.1
```

```
cat("4.4.1 ZEITAUFWAND PRO PERSON\n\n")
```

4.4.1 ZEITAUFWAND PRO PERSON

```
for (p in alle_personen) {
  sub <- all_data[all_data$person_id == p, ]
  if (nrow(sub) > 0) {
    cat(sprintf(
      "%s: decision= %.1f s  total= %.1f s  justif= %.1f s\n",
      p,
      round(mean(sub$decision_time_s, na.rm = TRUE), 1),
      round(mean(sub$total_time_s, na.rm = TRUE), 1),
      round(mean(sub$total_time_s - sub$decision_time_s,
        na.rm = TRUE), 1)
    ))
  }
}
```

```
A1: decision= 42.4 s  total= 122.4 s  justif= 80.1 s
A2: decision= 42.2 s  total= 123.4 s  justif= 81.1 s
A3: decision= 31.7 s  total= 106.8 s  justif= 75.1 s
B1: decision= 40.0 s  total= 100.7 s  justif= 60.8 s
B2: decision= 54.6 s  total= 104.4 s  justif= 49.8 s
B3: decision= 51.6 s  total= 99.6 s   justif= 48.0 s
C1: decision= 52.9 s  total= 154.8 s  justif= 101.9 s
C2: decision= 56.8 s  total= 126.8 s  justif= 70.0 s
C3: decision= 64.4 s  total= 122.3 s  justif= 57.9 s
```

```
cat("\nKI-Inferenzzeiten (zum Vergleich):\n")
```

KI-Inferenzzeiten (zum Vergleich):

```
cat(" Zero-Shot:", round(mean(zs$inference_time), 2), "s\n")
```

Zero-Shot: 4.74 s

```
cat(" LoRA:      ", round(mean(lora$inference_time), 2), "s\n")
```

LoRA: 7.57 s

K4.4.2: Entscheidungskonsistenz Methode: Fleiss' Kappa & Cohen's Kappa (Kapitel 3.5.3)
Grundlage: Tabelle 6 Fleiss' Kappa: gruppenspezifische Übereinstimmung (3 Rater) Cohen's
Kappa: paarweise Übereinstimmung (2 Rater)

```
cat("\n4.4.2 FLEISS' KAPPA PRO GRUPPEN\n")
```

4.4.2 FLEISS' KAPPA PRO GRUPPEN

```
mat_a <- as.matrix(pivot[, c("A1", "A2", "A3")])  
mat_b <- as.matrix(pivot[, c("B1", "B2", "B3")])  
mat_c <- as.matrix(pivot[, c("C1", "C2", "C3")])
```

```
k_a <- kappam.fleiss(mat_a)  
k_b <- kappam.fleiss(mat_b)  
k_c <- kappam.fleiss(mat_c)
```

```
cat("Gruppe A (keine KI):      kappa =",  
    round(k_a$value, 3), "\n")
```

Gruppe A (keine KI): kappa = 0.313

```
cat("Gruppe B (Zero-Shot-Modell): kappa =",  
    round(k_b$value, 3), "\n")
```

Gruppe B (Zero-Shot-Modell): kappa = 0.355

```
cat("Gruppe C (LoRA-Modell):      kappa =",  
    round(k_c$value, 3), "\n\n")
```

Gruppe C (LoRA-Modell): kappa = 0.151

```
cat("4.4.2 PAARWEISE COHEN'S KAPPA \n\n")
```

4.4.2 PAARWEISE COHEN'S KAPPA

```
paare <- list(
  c("A1","A2"), c("A1","A3"), c("A2","A3"),
  c("B1","B2"), c("B1","B3"), c("B2","B3"),
  c("C1","C2"), c("C1","C3"), c("C2","C3")
)

for (pair in paare) {
  k <- cohen.kappa(cbind(pivot[[pair[1]]], pivot[[pair[2]]]))
  cat(pair[1], "vs", pair[2],
      ": kappa =", round(k$kappa, 3), "\n")
}
```

```
A1 vs A2 : kappa = 0.303
A1 vs A3 : kappa = 0.139
A2 vs A3 : kappa = 0.51
B1 vs B2 : kappa = 0.396
B1 vs B3 : kappa = 0.248
B2 vs B3 : kappa = 0.44
C1 vs C2 : kappa = 0.156
C1 vs C3 : kappa = 0.184
C2 vs C3 : kappa = 0.177
```

K4.4.3: Übernahme von KI-Empfehlungen Methode: Übernahmeanalyse; Priorität binär kodiert (0/1) Begründung dreistufig (0 / 0,5 / 1)

```
cat("\n4.4.3 ÜBERNAHMERATEN: PRIORITÄT\n\n")
```

4.4.3 ÜBERNAHMERATEN: PRIORITÄT

```
bc <- all_data[all_data$group %in% c("B", "C"), ]

for (g in c("B", "C")) {
  sub <- bc[bc$group == g, ]
  cat(sprintf("Gruppe %s: %.3f\n", g,
              round(mean(sub$used_ai_priority, na.rm = TRUE), 3)))
}
```

Gruppe B: 0.533

Gruppe C: 0.167

```
cat("\n4.4.3 ÜBERNAHMERATEN: KRITERIEN \n\n")
```

4.4.3 ÜBERNAHMERATEN: KRITERIEN

```
kriterien_vars <- c("used_ai_business", "used_ai_urgency",  
                  "used_ai_impact",    "used_ai_risk",  
                  "used_ai_effort")  
  
kriterien_namen <- c("Business Value", "Urgency",  
                   "Impact", "Risk", "Estimated Effort")  
  
for (i in seq_along(kriterien_vars)) {  
  cat(kriterien_namen[i], ":\n")  
  for (g in c("B", "C")) {  
    sub <- bc[bc$group == g, ]  
    cat(sprintf("  Gruppe %s: %.3f\n", g,  
              round(mean(sub[[kriterien_vars[i]]],  
                        na.rm = TRUE), 3)))  
  }  
  cat("\n")  
}
```

Business Value :

Gruppe B: 0.667

Gruppe C: 0.267

Urgency :

Gruppe B: 0.650

Gruppe C: 0.333

Impact :

Gruppe B: 0.717

Gruppe C: 0.283

Risk :

Gruppe B: 0.683

Gruppe C: 0.300

Estimated Effort :

Gruppe B: 0.800

Gruppe C: 0.300

```
cat("4.4.3 ÜBERNAHME: BEGRÜNDUNG \n\n")
```

4.4.3 ÜBERNAHME: BEGRÜNDUNG

```
for (g in c("B", "C")) {  
  sub <- bc[bc$group == g, ]  
  cat(sprintf("Gruppe %s Verteilung (0 / 0,5 / 1):\n", g))  
  print(table(sub$used_ai_reasoning))  
  cat("Mittelwert:",  
      round(mean(sub$used_ai_reasoning, na.rm = TRUE), 3), "\n\n")  
}
```

Gruppe B Verteilung (0 / 0,5 / 1):

```
0 0.5 1  
12 19 29  
Mittelwert: 0.642
```

Gruppe C Verteilung (0 / 0,5 / 1):

```
0 0.5 1  
30 7 23  
Mittelwert: 0.442
```

K4.4.3: Übereinstimmung ohne explizite Übernahme: Wie viele Entscheidungen stimmten mit KI überein, obwohl die KI-Empfehlung nicht übernommen wurde?

```
bc$ki_priority <- NA  
bc$ki_priority[bc$group == "B"] <- zs$predicted_priority[  
  match(bc$story_id[bc$group == "B"], zs$key)]  
bc$ki_priority[bc$group == "C"] <- lora$predicted_priority[  
  match(bc$story_id[bc$group == "C"], lora$key)]  
  
bc$same_as_ki <- bc$priority == bc$ki_priority  
bc$agreed_not_adopted <- bc$same_as_ki & bc$used_ai_priority == 0
```

```

for (g in c("B", "C")) {
  sub <- bc[bc$group == g, ]
  cat(sprintf("Gruppe %s:\n", g))
  cat(" Gleich wie KI gesamt:           ",
      sum(sub$same_as_ki, na.rm = TRUE), "\n")
  cat(" Gleich & voll übernommen (1):  ",
      sum(sub$same_as_ki & sub$used_ai_priority == 1,
          na.rm = TRUE), "\n")
  cat(" Gleich & NICHT übernommen (0): ",
      sum(sub$agreed_not_adopted, na.rm = TRUE), "\n\n")
}

```

Gruppe B:

```

Gleich wie KI gesamt:           39
Gleich & voll übernommen (1):   32
Gleich & NICHT übernommen (0):  7

```

Gruppe C:

```

Gleich wie KI gesamt:           30
Gleich & voll übernommen (1):   10
Gleich & NICHT übernommen (0):  20

```

K4.6 + 4.6.1 + 4.6.2: Explorative Analyse: Demografische Einflüsse Methode: Demografische Analyse; Rein deskriptiv

```

for (p in alle_personen) {
  sub <- all_data[all_data$person_id == p, ]
  if (nrow(sub) > 0) {
    cat(sprintf(
      "%s | Rolle: %-16s | Erfahrung: %2s J. | Alter: %2s | Geschlecht: %-6s | Ø Priorität: %",
      p,
      unique(sub$role),
      unique(sub$experience),
      unique(sub$age),
      unique(sub$gender),
      round(mean(sub$priority_num, na.rm = TRUE), 2)
    ))
  }
}

```

A1 | Rolle: developer | Erfahrung: 25 J. | Alter: 50 | Geschlecht: male | Ø Priorität: 2.0

A2	Rolle: projectmanager	Erfahrung: 15 J.	Alter: 42	Geschlecht: male	Ø Priorität
A3	Rolle: other	Erfahrung: 3 J.	Alter: 25	Geschlecht: male	Ø Priorität
B1	Rolle: developer	Erfahrung: 4 J.	Alter: 24	Geschlecht: female	Ø Priorität
B2	Rolle: projectmanager	Erfahrung: 20 J.	Alter: 42	Geschlecht: female	Ø Priorität
B3	Rolle: other	Erfahrung: 5 J.	Alter: 27	Geschlecht: male	Ø Priorität
C1	Rolle: projectmanager	Erfahrung: 10 J.	Alter: 38	Geschlecht: male	Ø Priorität
C2	Rolle: developer	Erfahrung: 35 J.	Alter: 60	Geschlecht: male	Ø Priorität
C3	Rolle: other	Erfahrung: 8 J.	Alter: 31	Geschlecht: male	Ø Priorität

```
cat("\n--- Mittlere Priorität pro Rolle ---\n\n")
```

```
--- Mittlere Priorität pro Rolle ---
```

```
for (rolle in unique(all_data$role)) {
  sub <- all_data[all_data$role == rolle, ]
  cat(sprintf(
    "%-16s: Ø = %.2f  SD = %.2f  n = %d\n",
    rolle,
    round(mean(sub$priority_num, na.rm = TRUE), 2),
    round(sd(sub$priority_num, na.rm = TRUE), 2),
    nrow(sub)
  ))
}
```

```
developer      : Ø = 2.65  SD = 0.97  n = 60
projectmanager : Ø = 2.68  SD = 0.85  n = 60
other          : Ø = 2.68  SD = 1.03  n = 60
```

```
bc$jira_priority <- zs$original_priority[
  match(bc$story_id, zs$key)]

# Übernahmerate Priorität nach Jira-Ursprungspriorität
cat("\nÜBERNAHMERATE NACH JIRA-URSPRUNGSPRIORITÄT\n\n")
```

```
ÜBERNAHMERATE NACH JIRA-URSPRUNGSPRIORITÄT
```

```

for (g in c("B", "C")) {
  sub <- bc[bc$group == g, ]
  cat(sprintf("Gruppe %s:\n", g))
  res <- round(tapply(sub$used_ai_priority,
                      sub$jira_priority,
                      mean, na.rm = TRUE) * 100, 1)

  print(res)
  cat("\n")
}

```

Gruppe B:

Blocker	Critical	Major	Minor	Trivial
41.7	33.3	66.7	66.7	58.3

Gruppe C:

Blocker	Critical	Major	Minor	Trivial
0.0	16.7	25.0	25.0	16.7